

PROOF BY INDUCTION

Subtitle

E. Zimudzi and E. Thuma

Department of Computer Science
University of Botswana

20 October, 2025

OUTLINE

1. MATHEMATICAL INDUCTION

Mathematical Induction

To prove that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function, we complete two steps:

BASIS STEP: Verify that $P(1)$ is true.

INDUCTIVE STEP: Show that the conditional statement $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

To complete the inductive step, assume $P(k)$ is true for an arbitrary positive integer k and show that under this assumption, $P(k + 1)$ must also be true. The assumption that $P(k)$ is true is called the **inductive hypothesis**. Once complete both steps, we have shown that $P(n)$ is true for all positive integers n , that is, we have shown that $\forall n P(n)$ is true where the quantification is over the set of positive integers. In the inductive step, we show that $\forall k (P(k) \rightarrow P(k + 1))$ is true, where again, the domain is the set of positive integers. Expressed as a rule of inference, this proof technique can be stated as

$$(P(1) \wedge \forall k (P(k) \rightarrow P(k + 1))) \rightarrow \forall n P(n)$$

when the domain is the set of positive integers

Template for Proofs by Mathematical Induction

1. Express the statement that is to be proved in the form “for all $n \geq b$, $P(n)$ ” for a fixed integer b . For statements of the form “ $P(n)$ for all positive integers n ,” let $b = 1$, and for statements of the form “ $P(n)$ for all nonnegative integers n ” let $b = 0$. For some statements of the form $P(n)$, such as inequalities, you may need to determine the appropriate value of b by checking the truth values of $P(n)$ for small values of n .
2. Write out the words “Basis Step.” Then show that $P(b)$ is true, taking care that the correct value of b is used. This completes the first part of the proof.
3. Write out the words “Inductive Step” and state, and clearly identify, the inductive hypothesis, in the form “Assume that $P(k)$ is true for an arbitrary fixed integer $k \geq b$.”
4. State what needs to be proved under the assumption that the inductive hypothesis is true. That is, write out what $P(k + 1)$ says.

5. Prove the statement $P(k + 1)$ making use of the assumption $P(k)$. (Generally, this is the most difficult part of a mathematical induction proof. Decide on the most promising proof strategy and look ahead to see how to use the induction hypothesis to build your proof of the inductive step. Also, be sure that your proof is valid for all integers k with $k \geq b$, taking care that the proof works for small values of k , including $k = b$.)
6. Clearly identify the conclusion of the inductive step, such as by saying “This completes the inductive step.”
7. After completing the basis step and the inductive step, state the conclusion, namely, “By mathematical induction, $P(n)$ is true for all integers n with $n \geq b$.”

Example 1

Show that if n is a positive integer, then

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Solution: Let $P(n)$ be the proposition that the sum of the first n positive integers,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

We must do two things to prove that $P(n)$ is true for $n = 1, 2, 3, \dots$. Namely, we must show that $P(1)$ is true and that the conditional statement $P(k)$ implies $P(k+1)$ is true for $k = 1, 2, 3, \dots$.

BASIS STEP: $P(1)$ is true, $1 = \frac{1(1+1)}{2} = 1$. (ASIDE: The left-hand side of this equation is 1 because 1 is the sum of the first positive integer. The right-hand side is found by substituting 1 for n in $\frac{n(n+1)}{2}$.)

INDUCTIVE STEP:

For the inductive hypothesis we assume that $P(k)$ holds for an arbitrary positive integer k . That is, we assume that

$$1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}.$$

Under this assumption, we must prove $P(k) \rightarrow P(k+1)$, i.e. we must show that if

if $1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}$ then, $1 + 2 + 3 + \cdots + k + (k+1) = \frac{(k+1)((k+1)+1)}{2}$

$$1 + 2 + 3 + \cdots + k + (k+1) = \frac{(k+1)((k+1)+1)}{2} = \frac{(k+1)(k+2)}{2}, \text{ is true.}$$

Suppose $1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}$, then $1 + 2 + 3 + \cdots + k + (k+1) = \frac{(k+1)((k+1)+1)}{2}$.

$$\begin{aligned} 1 + 2 + 3 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$

This last equation shows that $P(k+1)$ is true under the assumption that $P(k)$ is true.

This completes the inductive step.

By mathematical induction we know that $P(n)$ is true for all positive integers n . That is, we have proven that

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2} \text{ for all positive integers } n.$$

Example 2

Show that for all nonnegative integers n ,

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1.$$

Solution: Let $P(n)$ be the proposition that

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1 \text{ for the integer } n.$$

BASIS STEP: $P(0)$ is true, because $2^0 = 1 = 2^1 - 1$.

INDUCTIVE STEP:

For the inductive hypothesis we assume that $P(k)$ holds for an arbitrary positive integer k . That is, we assume that

$$1 + 2 + 2^2 + \cdots + 2^k = 2^{k+1} - 1.$$

To carry out the inductive step using this assumption, we must show that when we assume that $P(k)$ is true, then $P(k+1)$ is also true. We must show that

$$1 + 2 + 2^2 + \cdots + 2^k + 2^{k+1} = 2^{k+2} - 1.$$

assuming the inductive hypothesis $P(k)$. Under the assumption of $P(k)$, we see that

$$\begin{aligned}
1 + 2 + 2^2 + \cdots + 2^k + 2^{k+1} &= (1 + 2 + 2^2 + \cdots + 2^k) + 2^{k+1} \\
&= 2^{k+1} - 1 + 2^{k+1} \\
&= 2 \cdot 2^{k+1} - 1 \\
&= 2^{k+2} - 1
\end{aligned}$$

Because we have completed the basis step and the inductive step, by mathematical induction we know that $P(n)$ is true for all nonnegative integers n . That is,

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1 \text{ for all nonnegative integers } n.$$

Example 3

Conjecture a formula for the sum of the first n positive odd integers. Then prove your conjecture using mathematical induction.

Solution: The sums of the first n positive odd integers for $n = 1, 2, 3, 4, 5$ are

$$1 = 1 \quad 1 + 3 = 4 \quad 1 + 3 + 5 = 9$$

$$1 + 3 + 5 + 7 = 16 \quad 1 + 3 + 5 + 7 + 9 = 25$$

n	sum of the first n odd natural numbers	n^2
1	$1 = \dots\dots\dots$	1
2	$1 + 3 = \dots\dots\dots$	4
3	$1 + 3 + 5 = \dots\dots\dots$	9
4	$1 + 3 + 5 + 7 = \dots\dots\dots$	16
5	$1 + 3 + 5 + 7 + 9 = \dots\dots\dots$	25
$\vdots$	$\vdots$	$\vdots$
n	$1 + 3 + 5 + 7 + 9 + 11 + \dots + (2n - 1) =$	n^2
$\vdots$	$\vdots$	$\vdots$

From these values it is reasonable to conjecture that the sum of the first n positive odd integers is n^2 , that is,

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

We need a method to prove that this conjecture is correct, if in fact it is.

Let $P(n)$ denote the proposition that the sum of the first n odd positive integers is n^2 .

Our conjecture is that $P(n)$ is true for all positive integers n . We first complete the basis step; that is, we must show that $P(1)$ is true.

Then we must carry out the inductive step; i.e. we must show that $P(k + 1)$ is true when $P(k)$ is assumed to be true.

BASIS STEP: $P(1)$ states that the sum of the first one odd positive integer is 1^2 . This is true because the sum of the first odd positive integer is 1. The basis step is complete.

INDUCTIVE STEP: To complete the inductive step we must show that the proposition $P(k) \rightarrow P(k+1)$ is true for every positive integer k . To do this, we first assume the inductive hypothesis. The inductive hypothesis is the statement that $P(k)$ is true for an arbitrary positive integer k , that is

$$1 + 3 + 5 + \cdots + 2k - 1 = k^2$$

To show that $\forall k(P(k) \rightarrow P(k+1))$ is true, we must show that if $P(k)$ is true (the inductive hypothesis), then $P(k+1)$ is true. Note that $P(k+1)$ is the statement that

$$1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1) = (k + 1)^2$$

$$\begin{aligned} 1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1) &= 1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1) \\ &= k^2 + (2k + 1) \\ &= k^2 + 2k + 1 \\ &= (k + 1)^2 \end{aligned}$$

This shows that $P(k+1)$ follows from $P(k)$. Note that we used the inductive hypothesis $P(k)$ in the second equality to replace the sum of the first k odd positive integers by k^2 .

The End